

TM Forum Introductory Guide

Autonomous Network Effectiveness Indicators

IG1256

Maturity Level: General availability (GA)	Team Approved Date:29-May-2026
Release Status: Pre-production	Approval Status: Team Approved
Version 3.2.0	IPR Mode: RAND

Notice

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1. Scope of Document

Autonomous Network effectiveness indicators gauge how *effective* your AN initiative is for your company or organization's business objectives. These Effectiveness Indicators complement the existing AN architecture (based on [IG1218](#) and [IG1230](#)) and the AN Levels Taxonomy that describes the characteristics of autonomy in operations. It describes the concepts, framework, design principles, and the reference set of autonomous network effectiveness indicators.

2. Background, Benefits, and Usage

2.1. Background and benefits

The recently published IG1326 "AN Empowering DT evolving towards level4" describes four elements key to any AN implementation, and effectiveness indicators are one of these key elements.

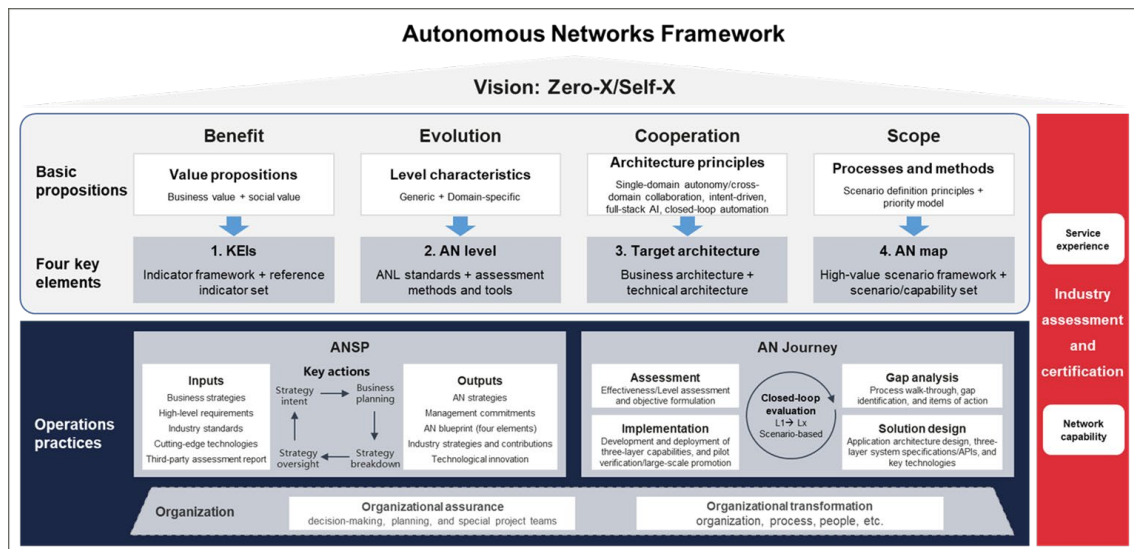


Figure 1. Four elements key to AN implementation

When an enterprise plans to undertake a strategy or investment, it must align its values and objectives within the organization and measure the return on investment. Central to this process is the definition and measurement of key indicators. The ultimate goal of Autonomous Networks is to enable the digital transformation and seamless service experience of vertical industries and consumers through Autonomous Networks/ICT services, meanwhile improving the operational efficiency of the telecom/ICT industry through automated, intelligent closed loop operations.

IG1252 defines an evaluation methodology for Autonomous Networks Levels (ANL). AN Levels gauge the level of autonomy capability in a given operational workflow or autonomous domain. However, AN Levels do not adequately reflect the overall impact or value of an autonomous network. Hence, there is a need to establish effectiveness indicators to help CSPs in determining the benefits of enhancing their telecommunications system with greater autonomy capabilities.

The value of these indicators lies in two aspects:

1. Visualizing and quantifying the effectiveness and benefits of AN evolution.
2. Aligning the development of autonomous capabilities with the enterprise strategy and service development trend.

Autonomous Network effectiveness indicators and ANL are two key AN evaluation factors that jointly facilitate the fulfillment of the autonomous network vision.

Autonomous Network effectiveness indicators can be used in the following three scenarios:

1. Formulating improvement objectives for AN based on effectiveness indicators. For example, if the service delivery target at L4 is measured in minutes, the target baseline can be set based on an operator's O&M status, experience of benchmark CSPs and annual investment budget. Note that the baseline value is not equivalent to a characteristic of L4.
2. Guiding capability development by associating effectiveness indicators with operation capabilities and converting and breaking down capability requirements for example, the time needed for optical transport network private line service delivery can be broken down into various key capability indicators, such as the time needed for CPE deployment and solution design. These capability indicators can be mapped to specific functional requirements for support systems, operations and maintenance centers, and network elements.
3. Verifying the effectiveness of operation capability development based on effectiveness indicators to provide inputs for the next AN Journey – the inputs include adjustments to the rationality of objectives, investment plan changes and optimized development directions.

2.2. Usage of this document

This document is intended for service providers and solution providers to evaluate the benefits of introducing autonomy capabilities into telecommunications systems and further identify the effectiveness of investments and specific areas for future improvement.

3. Definition of terms, and abbreviations

3.1. Terms

Term	Abbreviation	Description
Key Business Value Indicator	KBI	Indicator used for CSP C-Level Executives to evaluate the business value of introducing autonomous networks strategy.
Key Effectiveness Indicator	KEI	Indicator used for CSP Mid-Level Management to evaluate the outcome of introducing autonomy capability into telecom system in terms of business growth, customer experience, and operational efficiency.
Key Capability Indicator	KCI	Indicator used for autonomous network capability measurement and management.

For full list of terms related to autonomous networks, value stream etc. see IG1258, IG1230 and GB1040

4. Design Methodology of KEIs

4.1. AN Indicators and GB988 TM Forum Metrics

Many TMF groups have defined metrics in the TM Forum, and these are aggregated to form the well-known GB988. AN Project plans to contribute Autonomous Networks and Operation Metrics in GB988. Autonomous Networks and Operation Metrics consists of Key Business Value Indicators, Key Effectiveness Indicators and Key Capability Indicators.

GB988 TM Forum Metrics

BMS(Business Metrics Scorecard Team)	BSS(BSS Metrics)
CAB(Cable Team)	CEM(Customer Experience Management Team)
CF(Craig Farrell Book Metrics)	CYB(Cyberops Team)
DIG(Metrics added as part of digital overlay charter for Metrics team)	DMM(Digital Maturity Model)
FRD(Fraud Team)	MA(Margin Assurance)
MET(Central Metrics Team)	OMNI(Omni-B2B-CPQ Catalyst)
POST(Poster Completeness Metrics)	RA(Revenue Assurance Team)
REG(Regulatory)	SLAM(SLA Management)
VCPE(Virtual CPE)	WEB(Smart BPM enabling Digital Transformation)
WF(Workforce)	ZMPE(NFV ZOOM - ZOMP - ETSI)
ZMPQ(NFV ZOOM - ZOMP - QUEST)	Autonomous Networks and Operation Metrics

4.2. Autonomous Networks Effectiveness Indicators Framework

Based on the experiences gained in multiple CSPs' autonomous networks practices, we organize the autonomous networks and operation indicators in a framework is defined as shown in the following figure, Figure 2. The framework consists of three layers or hierarchies: Key Business Value Indicators Layer, Key Effectiveness Indicators Layer, and Key Capability Indicators Layer.

Key Business Value Indicators Layer: CSP executive formulate AN strategy. And develop key business value indicators (KBIs) based on AN value proposition. KBIs are used to set AN business value objective and demonstrate the business value of AN to executive.

Key Effectiveness Indicators Layer: Once the KBI goals are aligned within the CSP, goals are broken down to specific organizations and business units within the CSP, and KEI requirements for the organization and business are identified, tracked, managed, and measured. These indicators can be used to measure the service, organization, and operation improvements and benefits of CSPs after deployment autonomous network capabilities.

Key Capability Indicators Layer: CSP Key Effectiveness Indicators are further broken down to form requirements for AN capability. AN capability may be provided by the CSP or its suppliers. The capabilities can be measured by defining system, function, and feature indicators.

This document defines typical KBIs, KEIs, and KCIs to provide reference for AN measurement.

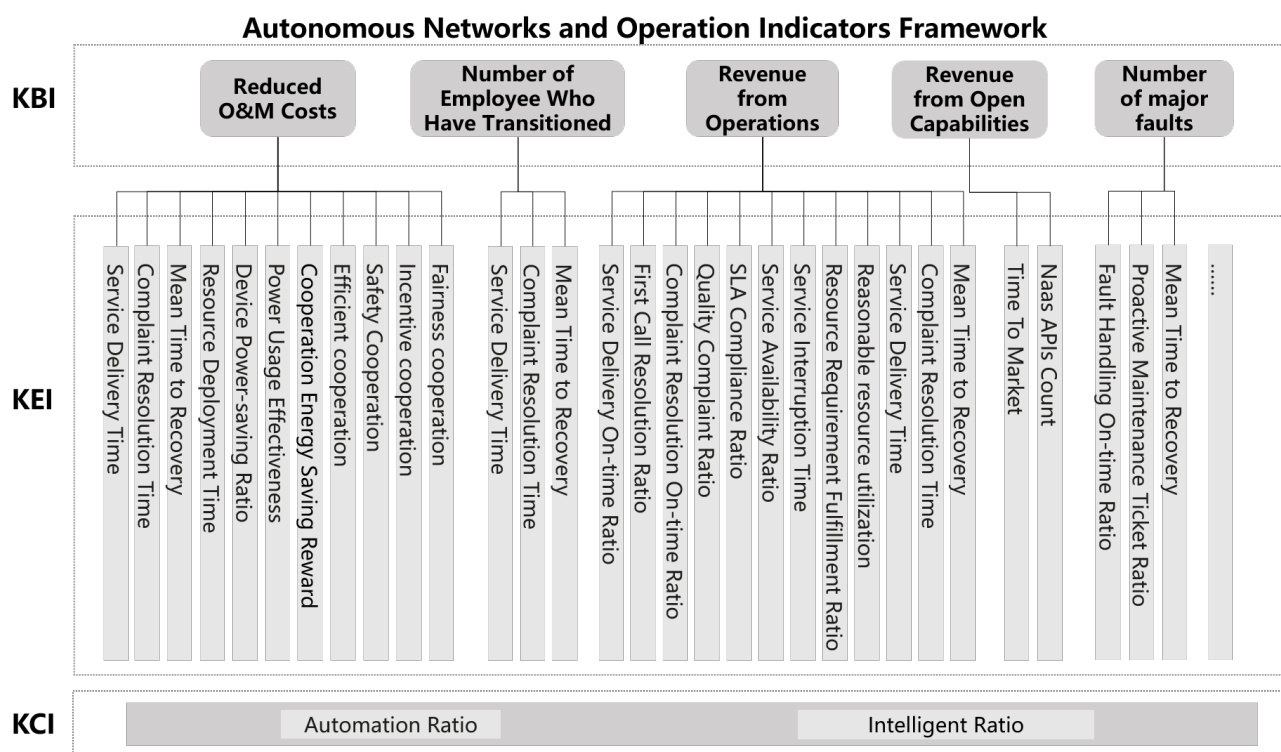


Figure 2. Autonomous Networks and Operation Indicators Framework

4.3. Autonomous Networks Effectiveness Indicators Design Principles

During the design of Autonomous Networks effectiveness indicators, we can refer to the SMART principles to clarify the key attributes of each indicator, such as the name, definition, description, formula, and form a set of Autonomous Networks effectiveness indicators with clear meaning, and comprehensive content. The overall design principles for effectiveness indicators are described as follows:

1. Specific:

The names and definitions of Autonomous Networks effectiveness indicators should not be too broad or vague. They should have clear context scenarios, boundaries, and measurement types (such as scale, degree, and ratio) to guide their implementation.

2. Measurable:

Autonomous Networks effectiveness indicators can be measured and evaluated quantitatively. That is, quantitative results can be calculated using specific formulas or rules based on a group of available original measurement data.

3. Achievable:

The implementation of Autonomous Networks effectiveness indicators is feasible. Specific steps can be formulated or converted into specific action requirements (such as automatic collection and calculation by the system). In addition, the definition of Autonomous Networks effectiveness indicators target values must be within the attainable scope. The benchmarks of other CSPs can be used to set reasonable achievement goals.

4. Relevance:

KBIs must be associated with value propositions. And each KEI should support at least one KBI value improvement. In addition, the association between KBIs and KEIs must be defined to form an indicators tree. This avoids indicator silos and facilitates the analysis of the impact of KEI changes on KBIs.

5. Time-bound:

Autonomous Networks effectiveness indicators are time-limited. That is, a reasonable statistical period needs to be set, for example, by month, quarter, or year, to facilitate historical comparison and trend analysis.

4.4. Autonomous Networks Effectiveness Indicators Definition Template

A template for the definition of KEI is shown below.

Name	Name of an Effectiveness Indicator.	Abbreviation	Abbreviation of an indicator
		Unit	Unit of an indicator. The common values are as follows: %: percentage #: numeral type \$: amount
Definition	Definition of indicator in one sentence.		
Description	Describe the value, function, and usage of the indicator.		
Calculation formula	Reference calculation formula for effectiveness indicator and necessary explanations and descriptions of related parameters in the formulas.		

5. KBI Reference Set

5.1. Reduced O&M Costs

Name	Reduced O&M Costs	Abbreviation	ROMC
		Unit	\$
Definition	Measure the reducing of O&M expenses achieved through deploy Autonomous Networks		
Description	This indicator reflects the expense reduction brought by introducing AN to the CSP's network, including the expense reduction in energy consumption (such as the electricity fee of the base station), improve the utilization of inefficient resources, and the reduction of outsourcing labor hours and so on.		
Calculation formula	$OMCS = \sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$ x_i Indicates the O&M cost saving value of type i. These types could be: • Savings on energy consumption expenses. • Savings from improved utilization of inefficient resources. • Savings from reducing outsourced staff hours, etc.		

5.2. Count of Employees Who Have Upskilled

Name	Count of Employees Who Have Upskilled	Abbreviation	COEWHU
		Unit	#
Definition	Measure the number of network operation and O&M personnel who have been upskilled.		
Description	The total time saved by the autonomous networks is converted to the number of maintenance personnel that can be upskilled.		
Calculation formula	$NOEWHU = \sum_{i=1}^n (t_i) \div$ $\text{Average annual working hours of an employee} = (t_1 + t_2 + \dots + t_n) \div \text{Average annual working hours of an employee}$ t_i indicates work time saved from work item i. Note: • 1. Work time saved: the total time saved varies according to the complexity, number of investments, and number of times of tasks. • 2. Average annual working hours of an employee: the actual investment of employees in different regions varies. • 3. Difficult in practice to calculate....		

5.3. Time Saved in Network Operations (Due to Employee Upskilling)

Name	Time Saved in Network Operations
Abbreviation	TSNO
Unit	Hours
Definition	Measures the time saved in network operations due to the implementation of autonomous functions, indicating improved efficiency.
Description	This metric reflects the network's efficiency by quantifying the reduction in operational time for various tasks (e.g., maintenance, monitoring, optimization) through autonomous capabilities. It provides a direct measurement of efficiency gains rather than the indirect result of workforce upskilling.
Calculation formula	TSNO = Total Manual Time – Total Autonomous Time

5.4. Revenue from Operations

Name	Revenue from Operations	Abbreviation	RFO
		Unit	\$
Definition	Measure the revenue growth brought by deploying an autonomous network to enhance the operation of existing services.		
Description	This indicator reflects the revenue growth through precision marketing, and service provisioning in advance, and loss of revenue decline caused by service SLA assurance and complaint handling time reduction		
Calculation formula	$RFO = \sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$ x_i indicates the revenue growth of type i. These types could be: • Increased revenue from early provisioning and early sales • Increased revenue from shorter interruption duration and less revenue loss • Increased revenue from experience improvement and stimulated traffic • Increased revenue from quality improvement and SLA monetization • Increased revenue from precision marketing/Sales		

5.5. Revenue from Open Capabilities

Name	Revenue from Open Capabilities	Abbreviation	RFOC
		Unit	\$
Definition	Measure the revenue generated by the open APIs to ecosystem partners.		
Description	This indicator reflects the value monetization results of carriers on open interfaces. Carriers have opened the following APIs to ecosystem partners: SMS verification code, entrusted payment, and location information. The more interfaces are provided, the greater the support for ecosystem partners is.		
Calculation formula	$RFOC = \sum_{i=1}^n (x_i \times k_i) = x_1 \times k_1 + x_2 \times k_2 + \dots + x_n \times k_n$ x_i indicates the number of times that type i open APIs are used. k_i indicates the unit price of type i open APIs. Note: 1. Open APIs refer to the interfaces opened by CSP to ecosystem partners, excluding the northbound interfaces used by NEs, OMCs, and OSSs. 2. The unit price refers to the average price for an ecosystem partner to invoke an OPEN API provided by CSP each time. The specific unit price is related to CSP's business model (for example, the more the API is invoked, the lower the unit price).		

5.6. Number of Major Faults

Name	Number of Major Faults	Abbreviation	NOMF
		Unit	#
Definition	Measure the number of major faults (such as network interruption, system breakdown, circuit blocking, charging failure, monitoring interruption, service interruption) that last for a certain period of time or affect more than a specified number of users.		
Description	This indicator reflects the network stability and security.		
Calculation formula	NOMF = Number of Major Service Faults + Number of Major Device Faults Major faults are classified into two types: major service faults and major device faults. Follow are references to some major faults. - Major service faults: 4G services (LTE data services are blocked and over XXXXX users are affected for more than XX minutes) - Home broadband service (more than XXXXX users in the same city are blocked for more than XX hour) - IoT service (more than XXXXXX subscribers under one APN are blocked for more than XX hour)		

Name	Number of Major Faults	Abbreviation	NOMF
		Unit	#
	- IMS service (The voice service of XXXXXX users in the province is interrupted for more than XX minutes) 2. Major network faults: - Transmission network (one WDM trunk is interrupted for X hours) - Wireless network (XX eNodeBs/gNodeBs are blocked for X hours) - Core network (STP breakdown or block for more than XX minutes) - Power and environment system (power failure and breakdown of the core equipment building)		

6. KEI Reference Set

6.1. KEIs related to Service/Network Monetization

6.1.1. Time To Market

Name	Time to Market	Abbreviation	TTM
		Unit	Day/Week/Month
Definition	Measure the time taken for new services or products from the initial concept stage to market launch.		
Description	This indicator reflects the ability of enterprises to launch new services or products. The shorter the time to market, the quicker a return on investment can be realized. AN capability such as self-orchestration and self-loading can quickly support the design, development, verification, and release of new services and products. The intent-based interface simplifies inter-system interconnection and shortens the integration time. All these AN capabilities can effectively shorten the product launch time.		
Calculation formula	TTM = Total time from concept to launch of all new services or products ÷ Number of new services or products		

6.1.2. NaaS APIs Count

Name	NaaS APIs Count	Abbreviation	NAC
		Unit	#
Definition	Number of open APIs opened by CSPs to ecosystem partners.		
Description	Network as a Service (NaaS) enables CSP to open network capabilities through APIs, explore new business models, and achieve business growth. Currently, CSPs have opened a few APIs to ecosystem partners, such as SMS authentication codes, entrusted payment, and location information. The more APIs are provided, the more support is provided for ecosystem partners.		
Calculation formula	NAC = Total number of NaaS open APIs Notes: The open APIs here refer to the APIs open to CSPs' ecosystem partners, excluding the northbound interfaces used internally.		

6.2. KEs related to Customer Experience

6.2.1. Service Delivery

6.2.1.1. Service Delivery Time

Name	Service Delivery Time	Abbreviation	SDT
		Unit	Minute/Hour/Day/Week/Month
Definition	Measure the average time required for delivering services to users.		
Description	This indicator reflects the efficiency of providing services to users. The smaller the value of this indicator, the faster the service is delivered and the shorter the waiting time for users. The average delivery time is applicable to service scenarios including but not limited to government and enterprise services, home broadband services, and individual services.		
Calculation formula	SDT = Total delivery time ÷ Number of Deliveries Note: The calculation of the delivery time can include the deployment of hardware or cable resources, or only the time required after the hardware resources are ready.		

6.2.1.2. Service Delivery On-time Ratio

Name	Service Delivery On-time Ratio	Abbreviation	SDOR
		Unit	%
Definition	Measure statistics on the proportion or services that are delivered on time.		
Description	This indicator reflects the efficiency of the operator's service delivery to users. Different from the service delivery time, service delivery on-time ratio is measured from the perspective of the proportion of delivered services that meet the requirements. Even if the service delivery time reaches the expected target, the service delivery on-time ratio may not meet the target because the delivery duration of each service may vary greatly. From the perspective of a single user, pay more attention to the service delivery time rather than the service delivery on-time ratio .		
Calculation formula	SDOR = (Number of instances that meet the service delivery time limit requirements ÷ Number of instances) × 100%		

6.2.2. Network Quality

6.2.2.1. SLA Compliance Ratio

Name	SLA Compliance Ratio	Abbreviation	SCR
		Unit	%
Definition	Measures the degree to which an CSP meets its SLA commitments.		
Description	This indicator reflects the CSP's ability to provide high-quality customer service and meeting its commitments with customers. The		

Name	SLA Compliance Ratio	Abbreviation	SCR
		Unit	%
	higher the rate, the better it suggests that the CSP is meeting the needs of customers and fulfilling promises.		
Calculation formula	$\text{SCR} = (\text{Number of instances met the SLA commitments} \div \text{Total number of instances}) \times 100\%$ Notes: SLA Refers to the service level commitment agreed in the business contract, such as availability requirements (e.g. ≥99.99%), latency requirements, bandwidth requirements, etc.		

6.2.2.2. Quality Complaint Ratio

Name	Quality Complaint Ratio	Abbreviation	QCR
		Unit	%
Definition	Measure the proportion of subscribers who complain about service and network quality.		
Description	This indicator reflects the impact of service and network quality problems on user experience. A higher complaint ratio has a greater impact on user experience and decreases user satisfaction.		
Calculation formula	$\text{QCR} = (\text{Number of complaints caused by network quality problems} \div \text{Total number of users}) \times 100\%$		

6.2.2.3. First Call Resolution Ratio

Name	First Call Resolution Ratio	Abbreviation	FCRR
		Unit	%
Definition	Measure the proportion of customer service agents who correctly address a customer's needs on the first call.		
Description	This indicator reflects whether the customer service agent can correctly meet the customer's requirements when the customer makes the first call without the need for a second call or contact.		
Calculation formula	$\text{FCR} = (\text{Number of times that the customer service agent can solve problems after the user makes the first call} \div \text{Total number of user calls}) \times 100\%$		

6.2.2.4. Complaint Resolution Time

Name	Complaint Resolution Time	Abbreviation	CRT
		Unit	Minute/Hour/Day
Definition	Measure the average time required for resolving a user complaint.		
Description	This indicator reflects how quickly and efficiently CSP can address their customers' problems and restore their trust. It is the average time it takes the customer service team from the moment a complaint is received to the time it is closed.		
Calculation formula	$\text{CRT} = \text{Total resolution time of all quality complaints} \div \text{Number of quality complaints}$		

Name	Complaint Resolution Time	Abbreviation	CRT
		Unit	Minute/Hour/Day
	Notes: Complaint handling includes the duration during which the customer cannot be reached due to the customer's reasons. This duration can be excluded from the measurement range.		

6.2.2.5. Complaint Resolution On-time Ratio

Name	Complaint Resolution On-time Ratio	Abbreviation	CROR
		Unit	%
Definition	Measure the proportion of complaints that are promptly resolved against quality issues reported by users to all complaints.		
Description	This indicator reflects the efficiency of operation personnel in handling customer complaints. The complaint resolution on-time ratio is measured from the perspective of the proportion of the complaint resolution time that reaches the target. The complaint resolution time reflects the average handling duration, even if the complaint resolution time reaches the expected target. The complaint resolution on-time ratio may not meet the requirement because the resolution time of each complaint may vary greatly. From the perspective of a single user, the complaint resolution time is more important than the complaint resolution on-time ratio.		
Calculation formula	$\text{SAR} = (1 - (\text{Service Unavailability Duration} - \text{Exemption duration}) \div \text{Service normal running duration specified in the contract}) \times 100\%$ Notes: Exemption duration refers to the planned service unavailability time agreed with the customer.		

6.2.2.6. Service Availability Ratio

Name	Service Availability Ratio	Abbreviation	SAR
		Unit	%
Definition	Measure the proportion of service uptime to total service time.		
Description	This indicator reflects the proactive prevention, prediction, quick response, and processing capabilities of CSPs in terms of service availability assurance. Generally, the customer will put forward clear requirements on service availability ratio, and the compensation clauses when the value does not meet the requirements.		
Calculation formula	$\text{SAR} = (1 - (\text{Service Unavailability Duration} - \text{Exemption duration}) \div \text{Service normal running duration specified in the contract}) \times 100\%$		

6.2.2.7. Service Interruption Time

Name	Service Interruption Time	Abbreviation	SIT
		Unit	Minute/Hour
Definition	Measure the total service interruption duration in the reporting time.		
Description	This indicator indicates the accumulated service interruption duration in the reporting period. A smaller value indicates a smaller negative impact on users. Different from the mean time to recovery, this counter		

Name	Service Interruption Time	Abbreviation	SIT
		Unit	Minute/Hour
	measures the total service interruption duration within a reporting period, not the duration of a single service interruption.		
Calculation formula	$\text{SIT} = (\text{Total services interruption time} - \text{Total exemption time}) \div \text{Total number of services}$ Notes: Exemption duration refers to the duration caused by failure or force majeure specified in the contract. It should be deducted from the service interruption duration.		

6.2.2.8. Mean Time to Recovery

Name	Mean Time to Recovery	Abbreviation	MTTR
		Unit	Minute
Definition	Measure the length of time between problem occurrence and recovery.		
Description	This indicator reflects the average recovery duration from the faulty state to the normal state. A smaller value indicates a faster recovery. The fault can be recovered by rectifying the fault point, or by other methods such as switchover and migration.		
Calculation formula	$\text{MTTR} = \text{Total recovery time} \div \text{Number of failures}$		

6.2.2.9. Fault Handling On-time Ratio

Name	Fault Handling On-time Ratio	Abbreviation	FHOR
		Unit	%
Definition	Measure statistics on the proportion of service faults that can be rectified in time to all faults.		
Description	This indicator reflects the fault handling efficiency. The difference between this counter and MTTR is that even if the MTTR reaches the expected target, the fault handling on-time ratio may not reach the target because the troubleshooting duration varies greatly in different cases. From the perspective of a single user, the fault recovery time is more important than the timeliness of troubleshooting.		
Calculation formula	$\text{FHOR} = (\text{Number of faults that meet the fault recovery time limit requirements} \div \text{Number of faults}) \times 100\%$		

6.2.2.10. Proactive Maintenance Ticket Ratio

Name	Proactive Maintenance Ticket Ratio	Abbreviation	PMTR
		Unit	%
Definition	Measure the proportion of tickets that proactively identify potential risks through preventive or predictive capabilities.		
Description	This indicator reflects the capability of proactively detecting and solving potential risks. It can solve problems before faults or complaints occur, reducing user experience and complaints. For example, technical measures, such as optical module aging and DGI hardware aging, can be used to predict and prevent hardware resources with high fault		

Name	Proactive Maintenance Ticket Ratio	Abbreviation	PMTR
		Unit	%
	frequency and poor service quality. The larger the value of this indicator, the better.		
Calculation formula	PMTR = (Number of potential trouble tickets identified by the prevention and prediction capability ÷ Number of trouble tickets) × 100% Notes: 1. To measure this indicator, the Operator shall provide trouble tickets and corresponding log information for a specified period (e.g., one month). 2. The system shall avoid duplicate ticket creation and ensure the accuracy of forecast tickets, to prevent wasted confirmation efforts arising from invalid tickets.		

6.2.2.11. Quality Optimization Duration

Name	Quality Optimization Duration	Abbreviation	QOD
		Unit	Minute/Hour
Definition	Measure the average time required for optimization when network quality issues occur.		
Description	This indicator reflects the efficiency of operators' activities in ensuring stable network quality and preventing and eliminating potential risks, covering phases such as quality degradation perception, deterioration prediction, demarcation and localization, optimization scheme development, evaluation and decision-making, as well as scheme implementation and verification.		
Calculation formula	QOD = (The total time taken from the creation to the closure of all network quality issue work orders within the statistical period ÷ Number of network quality issue work orders) × 100% Notes: 1. To measure this indicator, the operator shall provide optimization tickets or records covering a specified period (e.g., one month) 2. The calculation shall exclude any optimization records marked as failed due to insufficient bandwidth capacity. 3. For degradation scenarios caused by hardware failures requiring on-site repair, this indicator measures the time required to restore network or service quality, rather than the time spent on hardware fault rectification.		

6.2.2.12. Mean Time to Repair

Name	Mean Time to Repair	Abbreviation	MTTR
		Unit	Minute/Hour
Definition	Measure the time from the problem occurrence to repair.		

Name	Mean Time to Repair	Abbreviation	MTTR
		Unit	Minute/Hour
Description	This indicator reflects the average repair duration from the occurrence of a failure to the restoration of full functionality of the system or equipment. A smaller value indicates a faster repair. This duration includes the time taken for fault awareness, analysis, solution generation, and repair.		
Calculation formula	MTTR = Total repair time ÷ Number of faults		

6.2.2.13. Zero-touch Trouble Ticket Ratio

Name	Zero-touch Trouble Ticket Ratio	Abbreviation	ZTTTR
		Unit	%
Definition	Measure the percentage of trouble tickets without NOC manual intervention among all trouble tickets.		
Description	This indicator reflects the NOC zero-touch ratio of trouble tickets.		
Calculation formula	ZTTTR = Number of trouble tickets without NOC manual intervention / Number of trouble tickets within a specific period Notes: - To measure this indicator, the Operator shall provide trouble tickets and corresponding log information for a specified period (e.g., one month). - No manual intervention with NOC means that no NOC human employee effort on O&M activities such as correlation, root cause analysis, diagnosis, network configuration, etc. The activities exclude information review and decision approval.		

6.2.2.14. First On-site Resolution Ratio

Name	First On-site Resolution Ratio	Abbreviation	FOSRR
		Unit	%
Definition	Measures the ratio of tickets requiring on-site handling that are resolved on the first on-site visit.		
Description	This indicator reflects the on-site problem resolution effectiveness of Field Maintenance Engineers (FME). It adheres to the principle of "No on-site maintenance unless necessary" and stipulates that all necessary on-site maintenance tasks shall be resolved correctly on the first visit.		
Calculation formula	FOSRR = First On-site Resolution ticket / Total number of tickets requiring on-site handling × 100%		

6.3. KEs related to Sustainability

6.3.1. Energy saving

6.3.1.1. Device Power-saving Ratio

Name	Device Power-saving Ratio	Abbreviation	DPR
		Unit	%
Definition	Measure the power consumption decrease ratio before and after the power saving mode is enabled.		
Description	This indicator reflects the power saving capability of the device. A larger value indicates a better power saving effect. Energy saving must be implemented without affecting device performance and service experience, or the impact can be evaluated and controlled.		
Calculation formula	DPR = (1 - Power consumption when the power saving mode is enabled ÷ consumption when the power saving mode is disabled) × 100%		

6.3.1.2. Power Usage Effectiveness

Name	Power Usage Effectiveness	Abbreviation	PUE
		Unit	#
Definition	Measure the energy performance of the data center by calculating the ratio of the energy used as a whole as compared with the energy used by just the IT equipment alone.		
Description	This indicator reflects the energy efficiency of a data center. The total energy consumption of a data center includes the energy consumption of IT devices and the necessary energy consumption for maintaining the normal running environment of devices, including cooling, dehumidification, dust removal, monitoring, and ambient lighting. The closer this metric value is to 1, the less energy the non-IT devices consume, the better the energy efficiency level.		
Calculation formula	PUE = Total Data Center Energy Consumption ÷ Energy consumption of IT equipment		

6.3.2. Resource Utilization

6.3.2.1. Reasonable Resource Utility Ratio

Name	Reasonable Resource Utility Ratio	Abbreviation	RRUR
		Unit	%
Definition	Measures the proportion of resources that are in a reasonably utilized (not too low or too high) state.		

Name	Reasonable Resource Utility Ratio	Abbreviation	RRUR
		Unit	%
Description	This indicator reflects the efficiency of network resource usage. This indicator focuses on both underutilized resources and overutilized resources. Overutilized resources may reduce network reliability, interrupt services due to congestion or cause poor quality, and cause resource insufficiency during service delivery. Whether resources are heavily loaded or lightly loaded depends on the operator's expected usage of network resources in each domain. The specific range is subject to the carrier's resource usage management requirements. The method of calculating the number of resources varies according to the field. Take radio access network as an example, the resource is cell. That is, if the traffic of a cell covered by a wireless base station is within a threshold range during busy hours, the cell is considered as a reasonable used cell.		
Calculation formula	$RRUR = 1 - (\text{High-load resources} + \text{Low-load resources}) \div \text{Total resources} \times 100\%$		

6.3.2.2. Resource Requirement Fulfillment Ratio

Name	Resource Requirement Fulfillment Ratio	Abbreviation	RRFR
		Unit	%
Definition	Measure the proportion of existing resources that meet service delivery requests.		
Description	This indicator reflects whether resources on the live network provide timely and effective support for service delivery. It helps to reduce service delivery delays or even failures caused by resource unavailability.		
Calculation formula	$RRFR = (\text{Number of instances that meet resource requirements} \div \text{Number of instances}) \times 100\%$		

6.4. KEs related to Operation Efficiency

6.4.1. Process Efficiency

6.4.1.1. Resource Deployment time

Name	Resource Deployment Time	Abbreviation	RDT
		Unit	Minute/Hour/Day
Definition	Measures the time required for deploying the network resources.		
Description	This indicator reflects the efficiency of software resource deployment, including system check, software and configuration preparing, deployment implementation, commissioning, verification and acceptance tests, etc. This indicator does not involve the planning and design and hardware installation before		

Name	Resource Deployment Time	Abbreviation	RDT
		Unit	Minute/Hour/Day
	device power-on. After devices are installed and powered on, software resources must be deployed to provide services.		
Calculation formula	RDT = Deployed resource is ready to use – Deployment start time		

7. KCI Reference Set

7.1. Task Operation Automation Ratio

Name	Task Operation Automation Ratio	Abbreviation	TOAR
		Unit	%
Definition	Measures the proportion of subtasks that can be automatically completed in a type of task.		
Description	This indicator reflects the degree of automation of task execution. The higher the indicator, the greater the proportion of automated work, the less time human participation, and the higher the efficiency. The operations here refer to the activities related to the operation and maintenance, such as service delivery, complaint handling, fault handling, network optimization, and network planning/deployment. For example, the fault handling automation rate can be defined and measured based on the phased objective requirements. For example, the fault diagnosis automation rate. Operations that require on-site operations (such as hardware replacement) are not included in the statistics. For example, if hardware needs to be replaced and the system automatically provides a site visit solution (for example, replacing the XX board), the subtask can be considered as automatic.		
Calculation formula	$\text{OAR} = (\text{Number of subtasks that are automatically completed} \div \text{Total Number of subtasks}) \times 100\%$		

7.2. End-to-End Automation Ratio

Name	End-to-End automation ratio	Abbreviation	E2EAR
		Unit	%
Definition	Measure the proportion of end-to-end automated sub-scenarios of a value scenario.		
Description	This indicator reflects the automation ratio of sub-scenarios in high-value scenarios.		
Calculation formula	$\text{E2EAR} = (\text{Number of sub-scenarios that are automatically completed} \div \text{Total Number of sub-scenarios}) \times 100\%$ Note:		

Name	End-to-End automation ratio	Abbreviation	E2EAR
		Unit	%
	1. In addition to necessary manual onsite operations (such as onsite construction, hardware operation, and high-risk command review). Other steps should automatically and continuously process without manual intervention. 2. 2. Scenario: For details about value scenarios, see the definition of IG1339 and AN blueprint. Such as RAN energy saving and fault management and so on. 3. Sub-scenarios: Value scenarios are further divided into a series of sub-scenarios. For example, home broadband service complaints can be further classified into network disconnection, network disconnection, slow web page refreshing, and optical modem faults.		

7.3. Fault Management Automation Ratio

Name	Fault Management Automation Ratio	Abbreviation	FMAR
		Unit	%
Definition	Measure the proportion of trouble tickets that can be automatically managed by the system.		
Description	This indicator reflects the automatic management ratio for trouble tickets.		
Calculation formula	$\text{FMAR} = (\text{Number of trouble tickets fully automated by the system} + \text{Number of tickets with all processes automated except on-site engineering handling}) \div \text{Total number of trouble tickets} \times 100\%$ Note: To measure this indicator, the operator shall provide trouble tickets covering a specified period (e.g., one month).		

7.4. Automatic Fault Demarcation and Localization Ratio

Name	Automatic Fault Demarcation and Localization Ratio	Abbreviation	AFDLR
		Unit	%
Definition	Measure the proportion of trouble tickets where the system can automatically and accurately identify the root cause.		
Description	This indicator reflects the automatic fault demarcation and localization ratio of trouble tickets.		
Calculation formula	$\text{E2EAR} = (\text{Number of trouble tickets where the system can automatically and accurately identify the root cause} \div \text{Number of trouble tickets}) \times 100\%.$ Notes:		

Name	Automatic Fault Demarcation and Localization Ratio	Abbreviation	AFDLR
		Unit	%
	1. To measure this indicator, the Operator shall provide trouble tickets covering a specified period (e.g., one month). Each ticket shall record the ticket creation time, ticket diagnosis time, the system-automatically identified root cause, and the manually logged root cause category in an atomic manner. 2. The automatically identified root cause shall be consistent with the manually recorded root cause (i.e., the smallest field replaceable unit). 3. For automatic fault demarcation and localization, the time difference between Ticket Auto-Diagnosis Time and Ticket Creation Time shall not exceed a specified value (e.g., 5 minutes).		

7.5. Automatic Fault Rectification Ratio

Name	Automatic fault rectification ratio	Abbreviation	AFRR
		Unit	%
Definition	Measure the proportion of trouble tickets that can be automatically rectified by the system.		
Description	This indicator reflects the automatic fault rectification ratio of trouble tickets.		
Calculation formula	$\text{AFRR} = (\text{Number of faults automatically rectified by the system}) \div (\text{Total number of fault tickets} - \text{Number of tickets requiring on-site engineering handling}) \times 100\%$ Notes: 1. To measure this indicator, the Operator shall provide trouble tickets and corresponding log information for a specified period (e.g., one month). 2. For automatic rectification, all faults rectified within 15 minutes of occurrence shall be included in the statistics.		

7.6. Intelligent Ratio

Name	Intelligent ratio	Abbreviation	IR
		Unit	%
Definition	Measure the ratio of intelligent network infrastructure. An intelligent network infrastructure refers to an infrastructure configured with intelligent boards, intelligent modules, and intelligent protocols.		

Name	Intelligent ratio	Abbreviation	IR
		Unit	%
Description	Intelligent infrastructure is the basis for implementing autonomous network L4. Compared with traditional NEs, intelligent NEs have more accurate sensing capabilities, extending the sensing scope from active resources to passive resources. The perception precision is changed from port level to service level. (such as flow detection and poor service quality identification) The timeliness of perception is improved from hour-level and minute-level to second-level and sub-second-level.		
Calculation formula	IR =(Number of intelligent network infrastructure ÷ Total Number of network infrastructure) × 100%		

7.7. Cooperation

Based on the observation of four cooperation KCI (efficiency, safety, incentive-compatibility, and fairness) the effectiveness of the cooperation of the self negotiation services of a cooperation intent could be monitored.

7.7.1. Efficient Cooperation

Name	Efficient Cooperation	Abbreviation	EC
		Unit	[0,1]
Definition	Evaluate the efficiency of overall cooperation across episodes		
Description	After several iterations of cooperation step r involving N CSPs, each CSP agent i, acting on the behalf of one CSP and performing actions based on its policy cooperation $\hat{\pi}_i$, negotiates the on-guard service during T_{max} episodes. $G_i(\hat{\pi}, t)$ is the expected return, i.e. the sum of discounted rewards received by an agent i during one episode t (i.e., one negotiation) An efficiency property (E) measures how close the social welfare is from the optimum, which happens when all agents cooperate. The social welfare is the sum of expected returns over all CSPs agents. An optimal cooperating policy is noted $\pi_{i,C}$, whereas the worst cooperating policy is noted $\pi_{i,D}$.		
Calculation formula	$E(\hat{\pi}, r, t) = \frac{\sum_{i=1}^N G_i(\hat{\pi}_i, r, t) - \sum_{i=1}^N G_i(\pi_{i,D}, r, t)}{\sum_{i=1}^N G_i(\pi_{i,C}, r, t) - \sum_{i=1}^N G_i(\pi_{i,D}, r, t)}$		

7.7.1.1. Safety Cooperation

Name	Safety Cooperation	Abbreviation	SC
		Unit	[0,1]
Definition	Evaluate the efficiency of overall cooperation across episodes		
Description	After several iterations of cooperation step r involving N agents, each agent i , acting on the behalf of one CSP and performing actions based on		

Name	Safety Cooperation	Abbreviation	SC
		Unit	[0,1]
	its cooperation policy $\hat{\pi}_i$, negotiates the on-guard service during T_{max} episodes. $G_i(\hat{\pi}, t)$ is the expected return, i.e. the sum of discounted rewards received by an agent i during one episode t (i.e., one negotiation) A safety (Sf) property measures the risk taken by an agent i and is defined as the difference, when all other agents defect ($\pi_{-i,D} = \pi_j, j \neq i, D$), between the expected return received by agent i when trying to cooperate and the expected return it received when defecting. $(T - S)$ represents the maximal amplitude and, together with +1, allow normalizing Sf between 0.0 and 1.0		
Calculation formula	$Sf(\hat{\pi}_i, r, t) = \frac{G_i(\hat{\pi}_i, \pi_{-i,D}, r, t) - G_i(\pi_{i,D}, \pi_{-i,D}, r, t)}{T - S} + 1$		

7.7.2. Incentive Cooperation

Name	Incentive Cooperation	Abbreviation	IC
		Unit	[0,1]
Definition	Evaluate the efficiency of overall cooperation across episodes		
Description	After several iterations of cooperation step r involving N agents, each agent i acting on the behalf of one CSP and performing actions based on its cooperation policy $\hat{\pi}_i$, negotiates the on-guard service during T_{max} episodes. $G_i(\hat{\pi}, t)$ is the expected return, i.e. the sum of discounted rewards received by an agent i during one episode t (i.e., one negotiation) An incentive-compatibility property (IC) measures the capacity to incentivize cooperation and is defined as the difference, when all other agents try to cooperate ($\hat{\pi}_{-i} = \hat{\pi}_j, j \neq i$), between the expected return received by an agent i when trying to cooperate and the expected return it received when defecting. $(T - S)$ represents the maximal amplitude and allows normalizing IC.		
Calculation formula	$IC(\hat{\pi}_i, r, t) = \frac{G_i(\hat{\pi}_i, \hat{\pi}_{-i}, r, t) - G_i(\pi_{i,D}, \hat{\pi}_{-i}, r, t)}{T - S}$		

7.7.3. Fairness Cooperation

Name	Fairness Cooperation	Abbreviation	FC
		Unit	[0,1]
Definition	Evaluate the fairness efficiency of overall cooperation across episodes		
Description	After several iterations of cooperation step r involving N agents, each agent i acting on the behalf of one CSP and performing actions based on its cooperation policy $\hat{\pi}_i$, negotiates the on-guard service		

Name	Fairness Cooperation	Abbreviation	FC
		Unit	[0,1]
	during Tmax episodes. $G_i(\pi, t)$ is the expected return, i.e. the sum of discounted rewards received by an agent i during one episode t (i.e., one negotiation). Fairness property (J) with e_i equal to the total number of reward (e.g. kWh) saved by each agent i at the end of an episode t since the beginning of the evaluation after the several iterations of cooperation. In case of blacklisted agent(s), the Jain index is calculated based on the non-blacklisted agents.		
Calculation formula	$J(\vec{\pi}, r, t) = \frac{(\sum_{i=1}^N e_{i,t,r})^2}{N \sum_{i=1}^N e_{i,t,r}^2}$		

7.8. Adaptability

7.8.1. Time to Adapt

Name	Time to Adapt
Abbreviation	TTA
Unit	Seconds (s)
Definition	Measures the time taken for the Autonomous Network to begin its corrective response after detecting a change in the environment.
Description	This KCI captures the initial responsiveness of the AN. It starts when the AN detects a change (e.g., sudden increase in demand) and ends when the network initiates its adaptation processes to counteract the impact of the change. Faster "Time to Adapt" indicates high responsiveness and agility in managing dynamic conditions.
Calculation formula	$TTA = t_{start_response} - t_{change_detected}$

7.8.2. Time in Equilibrium

Name	Time in Equilibrium
Abbreviation	TIE
Unit	Seconds (s)
Definition	Measures the time it takes for the Autonomous Network to reach a stable operating state after initiating corrective actions.
Description	This metric assesses how quickly the AN can stabilize performance after responding to a detected change. It begins after the initial adaptation (captured by "Time to Adapt") and ends when the network reaches a

Name	Time in Equilibrium
	consistent performance level or steady state. Shorter "Time in Equilibrium" indicates better stability and control in response to dynamic network changes.
Calculation formula	$TIE = t_{steady_state} - t_{start_response}$

8. Autonomous Network KEIs

Example

8.1. STC Automatic 5GC Fault Demarcation to reduce MTTR

For operational excellence, STC introduces the concept of zero-touch fault handling and strategically applies increasing levels of automation across the analysis and decision stages, aiming to minimize human intervention and optimize fault resolution processes.

STC effectively shortens the MTTR of 5G Cloud Core faults by automatically detecting and demarcating frequent faults, automatically demarcating critical faults within 5 minutes, and locating NFV cross-layer faults within 15 minutes.

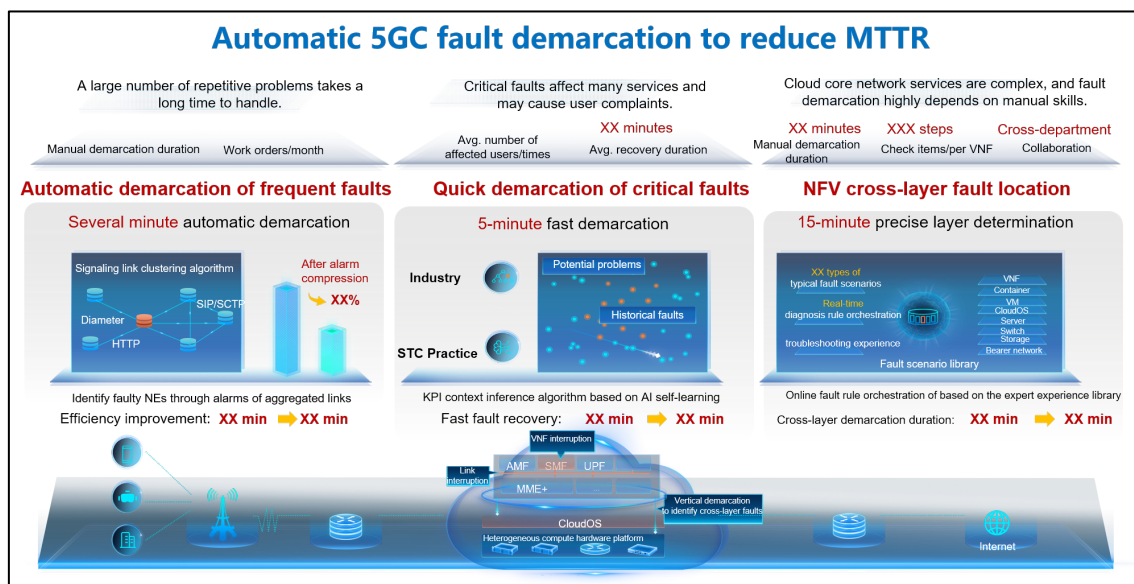


Figure 3. Automatic 5GC Fault Demarcation to reduce MTTR

8.2. Chunghwa Telecom Energy Saving for Optical/Copper Access Network

Chunghwa Telecom promotes energy conservation in fixed broadband network. The intelligent energy saving mechanism is implemented for a large number of broadband copper line access devices (more than 300,000 devices). Idle interfaces are disabled to reduce energy consumption. Automatic inventory is used for broadband optical access devices, idle power-consuming optical modules are identified, and power consumption is reduced through remote deactivation.

Through the intelligent energy saving mechanism, the annual power consumption of a single device is reduced from 20 kWh to 17 kWh, and the Device Power-saving Ratio is 15%.

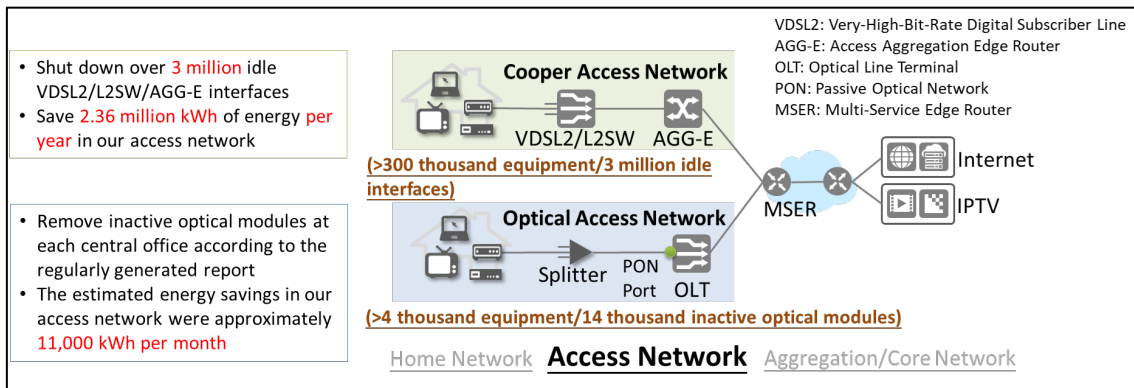


Figure 4. Energy Saving for Optical/Copper Access Network

8.3. Orange Inter-Operator Cooperation Scenarios to Save Radio Access Network Energy

To achieve zero carbon goals, CSPs must reduce their energy consumption sustainably and consistently. According to the GSMA, the telecommunications industry accounts for 2-3% of global energy consumption, and the majority (80%) is for Radio Access Networks (RAN). [Omdia]

Many competing CSPs have RANs that cover the same areas. As user traffic varies throughout the day, there are times (e.g., at night) when certain network features are generally unnecessary and can be disabled, but this is limited, as it must be done without degrading QoE for customers (e.g., turning off the power amplifier will also shut down connectivity) [Manson].

Cooperation between operators would allow an operator to periodically take turns connecting customers from different MNOs in a redundant coverage area during periods of network underload. The RANs of operators who do not stand guard in this area can be deactivated without degrading the QoE of the users.

"Turning off a RAN (2/3/4/5G) completely for certain periods could save a lot of energy, if it is feasible from a commercial and regulatory point of view." (Omdia) [Manson]. Rupert Wood and Caroline Gabriel "Driving down energy usage across telecoms networks: 5G RAN and beyond" Analysis Manson 2023 [Omdia] Guang Yang "RAN energy efficiency, key drivers, and areas of improvement" Omdia 2023

The use case shows how competing mobile operators (MNOs) can cooperate to reduce their energy consumption. It consists of designating the operator who will provide coverage within an area during certain periods, and redesigning the network according to the agreed sharing and division of responsibilities. This solution would allow significant energy savings, if it is commercially viable and authorized by the regulator, by switching off part or all of the radio access network (RAN) of the participating MNOs.

The objective is to minimize energy consumption (EC) and maximize energy efficiency (EE) among different mobile network operators (MNOs) using intelligence management components able to:

Self-negotiate agreements and decide on a fair division of responsibilities within areas with redundant network infrastructure with intelligent On guard tour (OGT) intent handler which apply cooperation intent.

Periodically self-redesign the networks based on the predicted network load and the self-negotiated deal (Intent update "new coverage area" decided by the On guard Tour (OGT)) identifying cells which could be switched off and how to adapt the rest of the network, including the core part with the Multi-criteria Intelligent Resource Allocation Intelligence management (MIRA).

Through the intelligence management (MIRA and OGT) energy could be saved up to 50%.

More information could be found in the white paper [AI-powered cooperation for efficient networks \(tmforum.org\)](https://tmforum.org/whitepapers/ai-powered-cooperation-for-efficient-networks)

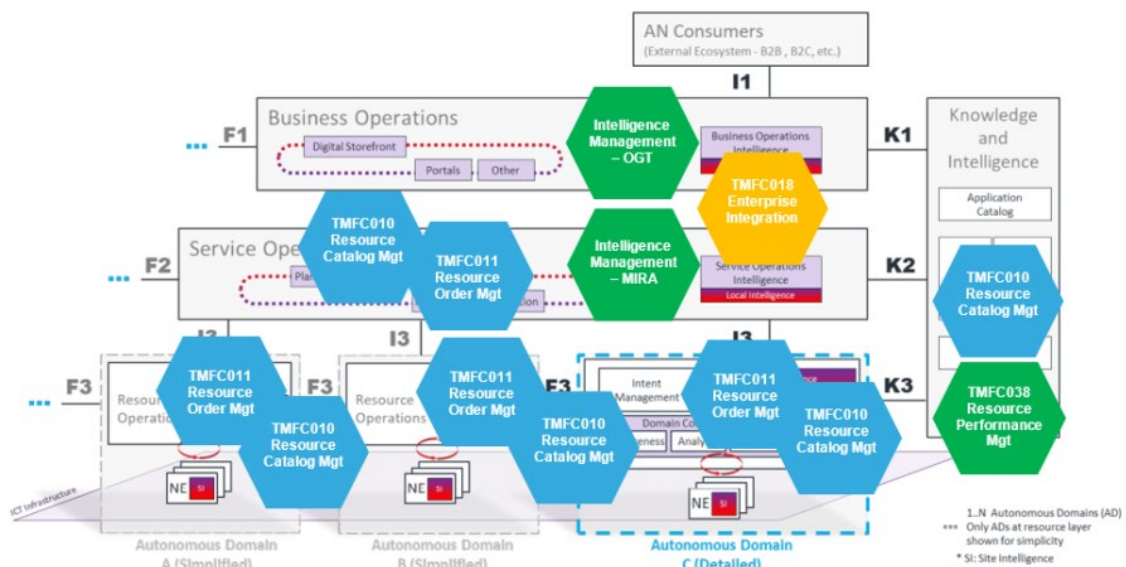


Figure 5. AI-powered cooperation for efficient networks

8.4. China Mobile Effectiveness Practice Referred to KEI Framework

China Mobile has been continuously deploying large-scale autonomous capability across the entire network in recent years, actively promoting the achievement of L4 of autonomous network.

Driven by business value and operational efficiency and referred to KEI framework, China Mobile has defined five value propositions, leading and promoting the transformation of network O&M from "device-oriented and management-oriented" to "customer-oriented, business-oriented, and service-oriented," aiming to achieve L4 advanced autonomous network by 2025.

Oriented to the value proposition, China Mobile specifies a framework of effectiveness indicators, including two classes of indicators as below.

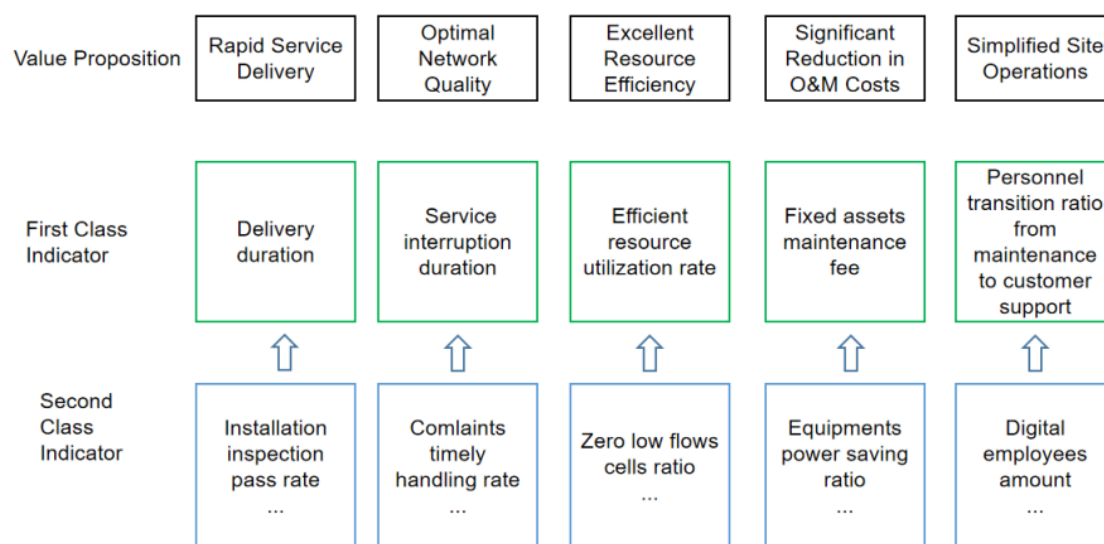


Figure 6. Framework of effectiveness indicators

Value Proposition

Rapid Service Delivery: Automatic network configuration, process visualized, end-to-end service delivery efficiency greatly improved.

Optimal Network Quality: End-to-end service quality assurance, customer satisfaction improved.

Excellent Resource Efficiency: Precise planning and rational use of network resources.

Significant Reduction in O&M Costs: O&M costs decreased year by year, equipments power efficiency continuously optimized for green and low-carbon.

Simplified Site Operations : basic maintenance efficiency greatly improved, and O&M personnel shift to customer support.

First Class Indicator

It is used to undertake the value proposition, which can be perceived and understood by senior managers and present AN overall value.

Second Class Indicator

Subdivision indicators for the specific O&M process of each network/service, supporting the achievement of first-class indicators, focusing on internal implementation.

Based on the above indicator framework and combined with the current domain scope of autonomous network, China Mobile further refines effectiveness indicator database and target baseline which will represents the effectiveness of key autonomy capabilities on specific network/service and forms a set of value evaluation method.

For example, China Mobile sets the automation rate of equipment inspection according to the Operation Automation Rate, which effectively reflects the O&M efficiency.

Name	Equipment Inspection Automation Rate for Core Network	Abbreviation	EIAR for CN
		Unit	%
Definition	Measures the proportion of equipment inspection for core network that can be automatically completed.		
Description	This indicator reflects the degree of automation of task execution. A higher automation rate of equipment inspection can improve inspection efficiency, reduce labor costs, reduce human errors, and improve the accuracy and reliability of inspection data. In addition, the automation rate of equipment inspection can also improve the reliability and stability of the equipment, reduce the failure rate of the equipment, and improve the service life of the equipment.		
Calculation formula	EIAR for CN = (Number of equipments inspection that are automatically completed ÷ Total Number of equipments inspection) × 100%		

8.5. AIS CODC Effectiveness Practice

AIS is a leading operator, with a strong commitment to providing a distinctive customer experience through Autonomous Networks. Since 2022, AIS has initiated the AN journey in value scenarios such as fault management and customer complaints. The company has designed an Effectiveness Indicators framework to quantify the effectiveness and benefits of AN evolution and to guide the realization of the benefits of innovation practices.

Cell outages are a common occurrence in Thailand, with a notable negative effect on customer experience. AIS has introduced the Cell Outage Detection and Compensation (CODC) feature, which enables real-time detection and automatic traffic compensation for outage cells.

Referring to "Service Interruption Time" in the standard, and considering the actual situation of the new feature, AIS defined Service Impact Time and further extended it to Traffic Lost Protection.

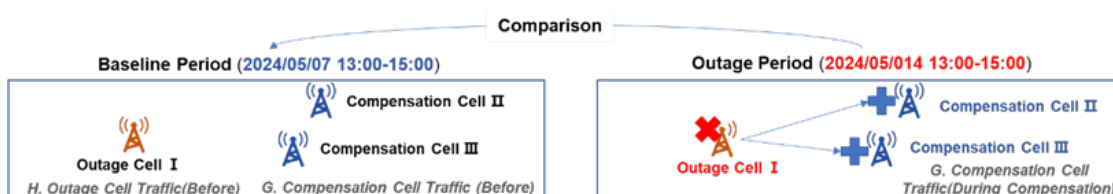


Figure 7. Cell Outage and Comparison

In the event of a cell outage, the antenna parameters of neighboring sites can be automatically adjusted in order to ensure uninterrupted service. From example picture of right-hand side, when Cell I experienced outage (red color), Cell II and Cell III can compensate (blue color) for a certain percentage of the original traffic on Cell I. And picture of left-hand side, it will collect traffic of Cell I, Cell II and Cell III of previous week with same period of hour for calculate baseline period.

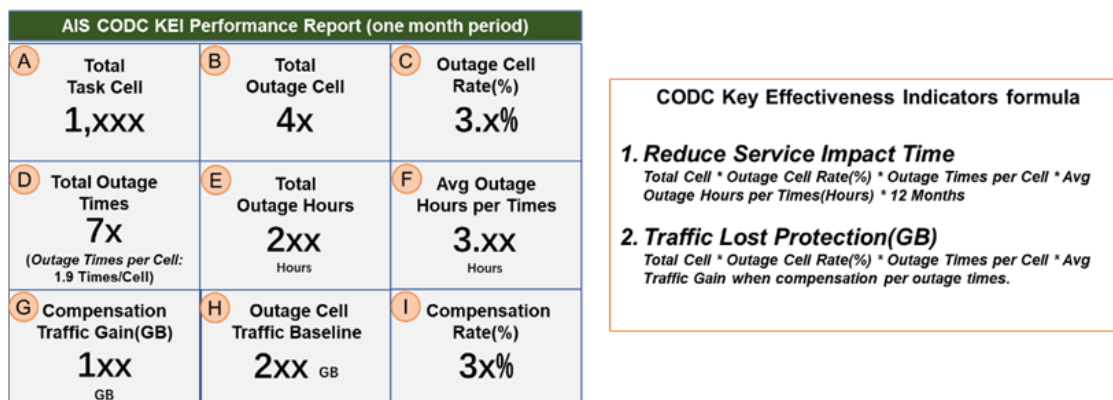


Figure 8. CODC KEI Report

AIS started to monitor the cell outage data of 1xxx task cells (A) over a one-month period following the introduction of the CODC feature. The results showed that 4x task cells experienced outages (B), indicating a cell outage rate of approximately 3.x% ($C = A/B$). The average outage times per cell (D') is calculated by dividing total number of cell outages (D) by total outage cells (B), and the average outage hours per times (F) is calculated by dividing the total outage hours (E) by total number of outages (D). Accordingly, the total cell outage hours of one year, which is referred to as the **service impact time**, can be calculated by multiplying the total number of cells, the cell outage rate (B), the average outage times per cell (D') and the average duration of outages (F). Reducing service impact hours is a key objective in our efforts to enhance customer experience.

The key effective indicator stat from the comparison of the same time period in the week before and after incidents happened. By dividing the traffic change of the compensation cells (G) by the traffic baseline of the outage cell (H), it is estimated that the CODC function achieves a compensation rate of 3x% (I). On this basis, it is also possible to calculate the **traffic lost protection** of total cells in a year by multiplying the total outage cells, the average outage times per cell (D') and average traffic gain of compensation per outage times

.

Name	Service Impact Time	Abbreviation	SIT
		Unit	Hour
Definition	This indicator measures the total duration for which services were affected during the reporting period.		
Description	This indicator reflects the cumulative duration of the impact of the service during the reporting period. Smaller values indicate less negative impact on users. This indicator differs from service disruption time mainly in that the affected service may not be disrupted.		
Calculation formula	SIT = Total Cell × Outage Cell Rate (%) × Outage Times per Cell × Avg Outage Hours per Times (Hours) × 12 Months		

Name	Traffic Lost Protection	Abbreviation	TLP
		Unit	GB
Definition	This indicator quantifies the amount of traffic that is protected during cell outages.		
Description	This indicator assesses the capability of maintaining service continuity when a cell outage occurs. It is measured by comparing the compensation traffic during the outage period with the baseline traffic. Larger values indicate less negative impact on users.		
Calculation formula	TLP = Total Cell × Outage Cell Rate (%) × Outage Times per Cell × Avg Traffic Gain when compensation per outage times.		

9. References

#	Title	Organization
IG1218	IG1218 Autonomous Networks Business Requirements & Architecture v2.0.0	TM Forum
IG1230	Autonomous Networks Technical Architecture V1.0.0	TM Forum
IG1252	IG1252 Autonomous Network Levels Evaluation Methodology v1.1.0	TM Forum
GB988	TM Forum Metrics Definition v21.5.0	TM Forum

10. Administrative

10.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1 - 0.8	15-Oct-2022	Wu Yunhe	Initial draft, several iterations
0.9	08-Nov-2022	Wu Yunhe	Edits after TA meeting held on 01-Nov-2022. Initial draft of stand-alone document on effectiveness indicators
0.91	22- Nov -2022	Wu Yunhe	Applying minor updates
1.0.0	09-Dec-2022	Alan Pope	Final edits prior to publication
1.1.0	15-Aug-2023	Wu Yunhe	Update based on the AN practices of CSP and supplier, summary value propositions and KPIs.
2.0.0	17-Oct-2023	Alan Pope	Final edits prior to publication
2.1.0	19-Dec-2023	Alan Pope	Final edits prior to publication
2.2.0	03-May-2024	Alan Pope	Final edits prior to publication
3.0.0	08-Nov-2024	Alan Pope	Final edits prior to publication
3.1.0	19-Sep-2025	Alan Pope	Final edits prior to publication
3.2.0	29-May-2026	Benson Yap	Final edits prior to publication

10.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	09-Dec-2022	Alan Pope	Initial release
Pre-production	23-Jan-2023	Adrienne Walcott	Updated to reflect Member Evaluated status
Pre-production	15-Aug-2023	Alan Pope	Updated to version 1.1.0.
Pre-production	18-Sep-2023	Adrienne Walcott	Updated to reflect Member Evaluated status
Pre-production	17-Oct-2023	Alan Pope	Updated to version 2.0.0
Production	08-Dec-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	19-Dec-2023	Alan Pope	Updated to version 2.1.0
Production	09-Feb-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	03-May-2024	Alan Pope	Updated to version 2.2.0
Production	28-Jun-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	08-Nov-2024	Alan Pope	Updated to version 3.0.0
Production	27-Dec-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	19-Sep-2025	Alan Pope	Updated to version 3.1.0
Production	14-Nov-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	29-May2026	Benson Yap	Updated to version 3.2.0

10.3. Acknowledgements

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